

Initialization

Data import

In[327]:=

```
fnd = FileNames["dls*.txt"];
TableForm[%]

BSA and apo-ferritin
```

In[329]:=

```
dlsmd[fnd[[4]]][[1]] // TableForm
dbsaa = Table[N@{10-6 dlsmd[fnd[[4]]][[4, i], dlsmd[fnd[[4]]][[5, i]]T, {i, 4}}];
dbsa = dbsaa[[1]];
dapf = dbsaa[[4]];
```

Out[329]//TableForm=

```
bsa 99   gf 3 mg/mL 1
hepes a #1 1
hepes a #1 filtered 20 nm 1
apo-ferritini 2 mg-mL 1
```

Polystyrene 60 nm

In[]:=

```
dlsmd[fnd[[5]]][[1]] // TableForm
dp60r = Table[N@{10-6 dlsmd[fnd[[5]]][[4, i], dlsmd[fnd[[5]]][[5, i]]T, {i, 3}}];
dp60 = dp60r[[1]];
```

Out[]//TableForm=

```
Nanosphere 60 nm 1
Nanosphere 60 nm 2
Nanosphere 60 nm 3
```

Converting $g_2 \rightarrow g_1$

g2 - 1 to g1 experimental data conversion

In[]:=

```
g1a = {xySc[dp60r[[1]], Identity,  $\sqrt{\frac{\#}{\text{Max}[dp60r[[1]][[All, 2]]}}$  &],
        xySc[dbsa, Identity,  $\sqrt{\frac{\#}{\text{Max}[dbsa[[All, 2]]}}$  &]]};
```

In[]:=

```
g1fit = a /. FindFit[#, Exp[-a t], {a}, t] & /@ g1a;
g1af = Table[xySc[g1a[[i]], # g1fit[[i]] &], {i, Length[g1a]};
Autocorrelation function  $e^{-t}$ 
```

In[]:=

```
expm = N@{#, Exp[-1 #]} & /@ Range[0, 6, 6];
The positions of fit values
```

In[]:=

```
tauval = Table[First@Nearest[g1a[[i], All, 1], g1fit-1[[i]]], {i, 2}];
taupos = Flatten[Table[Position[g1a[[i], All, 1], tauval[[i]]], {i, 2}]];
```

Visualization

```

In[ ]:= linleg =
  LineLegend[linstl[{Black, StandardRed}, None, 0.6], {"NIST 60(2) nm", "BSA"},
  LegendLabel → Style["Samples", Bold], LabelStyle → fs,
  LegendMarkers → {{sym[[1]], fs}, {sym[[2]], fs}}, LegendMarkerSize → {20, 5}];

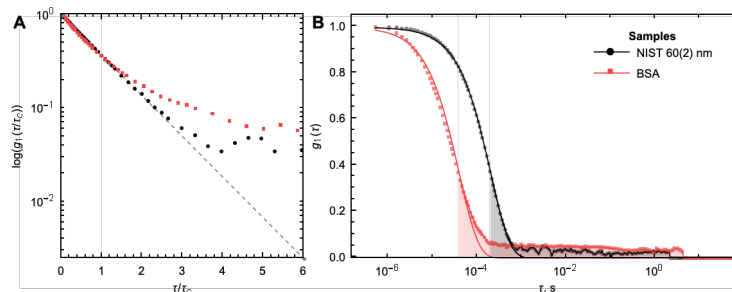
In[ ]:= gldimpl = ListPlot[Append[Table[Take[g1af[[i]], {66, 47}[[i]]], {i, 2}], expm],
  ScalingFunctions → "Log10", PlotMarkers → {Automatic, 4},
  GridLines → {{1}, None}, Joined → {False, False, True},
  PlotStyle → linstl[{Black, StandardRed, Gray}, {None, None, dotted}, 0.6],
  PlotRange → {{0, 6}, {expm[[1], 2], 1.0}}, ImagePadding → {{30, 5}, {25, 5}},
  ImageSize → is GoldenRatio-1, AspectRatio → 1,
  Epilog → {labels2[" $\tau/\tau_c$ ", "log( $g_1(\tau/\tau_c)$ )", "", {0.12, 0.18}, fs],
  Text[Style["t/ $\tau_c$  = 1", fs, White], Scaled[{0.1, 0.5}]]}, Evaluate[style]];

gltail = Show[ListPlot[g1a, ScalingFunctions → {"Log10"},
  PlotStyle → linstl[{Gray, Lighter@StandardRed}, None, 0.6],
  GridLines → {g1fit-1, None}, PlotMarkers → {Automatic, 4}],
  ListLinePlot[Table[Drop[g1a[[i]], taupos[[i]]], {i, 2}], ScalingFunctions →
  {"Log10"}, PlotStyle → linstl[{Black, StandardRed}, None, 0.6],
  GridLines → {g1fit-1, None}, Filling → Axis, PlotRange → All],
  Plot[Evaluate[Table[Exp[- g1fit[[i]] t], {i, 2}]], {t, 5 × 10-7, 3},
  PlotStyle → linstl[{Black, StandardRed}, None, 0.6], PlotRange → All,
  ScalingFunctions → {"Log10"}], PlotRange → {-0.005, 1.05}, ImageSize → is,
  AspectRatio → 1 / GoldenRatio, ImagePadding → {{30, 5}, {25, 5}},
  Epilog → {labels2[" $\tau$ ", s", " $g_1(\tau)$ ", "", {0.12, 0.09}, fs],
  Inset[linleg, Scaled[{0.77, 0.83}]]}, Evaluate[style]];

In[ ]:= Graphics[{White, EdgeForm[None], Rectangle[{2, 3}, {131, 52}],
  Inset[gldimpl, {27, 27}, Automatic, 56], Inset[gltail, {93, 27}, Automatic, 83],
  Text[Style["A", Bold, 1.5 fs, Black], {2, 51}],
  Text[Style["B", Bold, 1.5 fs, Black], {56, 51}]], ImageSize → 1.6 is,
  ImagePadding → {{0, 0}, {0, 0}}, PlotRangeClipping → None]
Export["/Users/heliosight/Documents/work/UmU/projects/06_coagulation/figures/
  dls_poly.pdf", %];

```

Out[]:=



Synthetic power law sample

Modeling the power law distribution g_1 autocorrelation functions for the polydispersity explanation plot.

```

In[ ]:= (*physical constants*)
kB = 1.380649 * 10^-23;
T = 296;
eta = 0.001;
(*optical geometry*)
lambda0 = 633 * 10^-9;
nref = 1.33;
theta = 173 Degree;
qvec = (4  $\pi$  nref / lambda0) Sin[theta / 2];
(*monomer radius*)
R0 = 3 * 10^-9;
(*aggregation numbers*)
Ns = Range[1, 100];
(*hydrodynamic radius*)
Rh[N_] := R0 N^(1 / 3);
(*diffusion coefficient*)
Dcoef[N_] := kB T / (6  $\pi$  eta Rh[N]);
(*DLS decay rate*)
rate[N_] := Dcoef[N] qvec^2;
(*power-law weights~N^-1.5*)
weightsr = Ns^-1.5;
(*normalize*)
weights = weightsr / Total[weightsr];
(*weighted decay function of ensemble*)
g1Total[tau_] := Total@Table[weights[[i]] Exp[-rate[Ns[[i]] tau], {i, Length[Ns]}}];
(*mean value of the time*)
meanTau = Total[weights / (rate /@ Ns)^2] / Total[weights / (rate /@ Ns)];

```

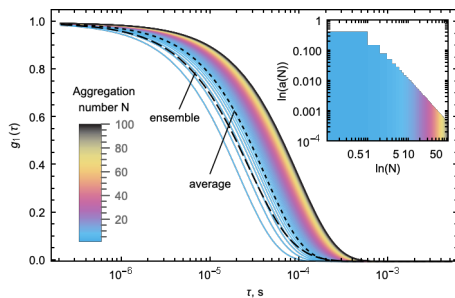
Illustrating the inversion problem and Jensen inequality

```

In[ ]:= colors = ColorData["CMYKColors"] /@ Rescale[Ns];
rcch = RectangleChart[ConstantArray[1, Length[Ns]], weights]^T,
  ScalingFunctions -> {"Log10", "Log10"}, PlotRange -> {{0.101, 99}, {10^-4, 1}},
  ChartStyle -> colors, impd2, AspectRatio -> 1, ImageSize ->  $\frac{1}{2.6}$  is,
  Epilog -> labels2["ln(N)", "ln(a(N))", "", {0.22, 0.37}, fs], Evaluate[style]];
blgnd = BarLegend[{"CMYKColors", {Min[Ns], Max[Ns]}], LegendLabel ->
  "Aggregation\!number N", LabelStyle -> fs, LegendMarkerSize -> {15, 80}];
encrd = {{Log[10, 6.5 \times 10^-6], 0.79}, {Log[10, 4.33 \times 10^-6], 0.64}};
avcrd = {{Log[10, 0.000018], 0.63}, {Log[10, 9.6 \times 10^-6], 0.368}};
polyth = Show[Plot[Evaluate@Table[Exp[-rate[N] tau], {N, Ns}],
  {tau, 0.2 \times 10^-6, 0.5 \times 10^-2}, ScalingFunctions -> {"Log10"},
  PlotRange -> {0, 1}, PlotStyle -> linstl[colors, None, 0.6]],
  Plot[{g1Total[tau], Exp[-tau/meanTau]}, {tau, 0.2 \times 10^-6, 0.5 \times 10^-2},
  ScalingFunctions -> {"Log10"}, PlotRange -> All, AxesLabel -> {"\tau (s)", "g_1(\tau)"},
  PlotStyle -> linstl[Black, {dashed, dotted}, 0.8]], impd2,
  ImageSize -> is, AspectRatio -> 1/GoldenRatio, PlotRange -> {-0.005, 1.05},
  Epilog -> {labels2["\tau, s", "g_1(\tau)", "", {0.12, 0.09}, fs],
  Inset[blgnd, Scaled[{0.13, 0.33}]], Inset[rcch, Scaled[{0.78, 0.64}]],
  {AbsoluteThickness[0.6], Line[encrd]}, Text[Style["ensemble", fs],
  {Log[10, 0.4 \times 10^-5], 0.6}], {AbsoluteThickness[0.6], Line[avcrd]},
  Text[Style["average", fs], {Log[10, 10^-5], 0.32}]], Evaluate[style]]
Export[
  "/Users/heliosight/Documents/work/UmU/projects/06_coagulation/figures/dls_poly
  _th.pdf", polyth, ImageResolution -> 600]

```

Out[]:=



Out[]:=

```

/Users/heliosight/Documents/work/UmU/projects/06_coagulation/figures/dls_poly
_th.pdf

```

The time points are taken from the actual Malvern g_1 function of BSA sample, to generate the realistic g_1 function as if measured by Nano S instrument. This section depends on **Data import** and **Synthetic power law sample**.

Noise model

Standard deviation forms according to the literature. They have a disadvantage of being monotonic

functions of data, which breaks the $\chi^2 \sim M$ condition, since the distribution of the errors is not normal as seen from the next plot in this section.

APPENDIX 7.B. ANOTHER PROPOSED FORM FOR σ_i^2

For a Poisson profile,

$$\sigma_i^2 = \frac{1 + d_i^2}{4Ad_i^2}.$$

On closer examination, PCS is seen to be a measure of deviations of Poisson distributions. Thus, statistical fluctuations should be included in the error

deviations. For example, the statistical error can be calculated by considering the signals, which obey a Gaussian distribution. Then σ_i^2 has two features. It is a constant according to the Poisson distribution, but it fluctuates at each data point when compared with the measured data. One proposed form is

$$\sigma_i^2 = \frac{1}{4A} d_i^{0.05}, \quad (7.B.1)$$

which is by no means unique. Its use instead of Eq. (7.6.14) improves the resolution of the MEM. Preliminary simulated tests show that the improved MEM can resolve two narrow (δ) bimodal distributions with $\bar{\Gamma}_1/\bar{\Gamma}_2 \lesssim 2$. Further improvements on the form of σ_i^2 are possible and should be forthcoming.

References

- [1] B. Chu. Laser light scattering : basic principles and practice. Academic Press, Boston, 2nd ed. edition, 1991.

```

In[ ]:= {gm, nm1, nm2} = Table[{e,  $\frac{1 + e^2}{e^2}$ ,  $e^{0.05}$ }, {e, 0.01, 1, 0.01}]^T;

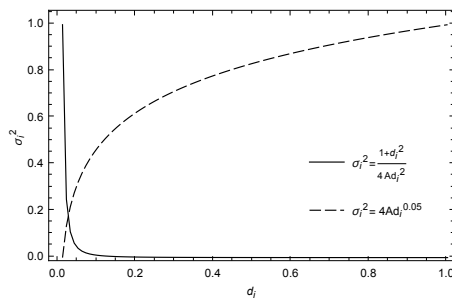
legErr = LineLegend[linstl[Black, {None, dashed}, 0.6],

  {" $\sigma_i^2 = \frac{1 + d_i^2}{4 A d_i^2}$ ", " $\sigma_i^2 = 4 A d_i^{0.05}$ "}, LabelStyle → fs, LegendMarkerSize → {20, 20}];

ListLinePlot[Evaluate[{gm, Rescale[#]^T & /@ {nm1, nm2}}],
  ScalingFunctions → None, ImageSize → is, AspectRatio → 1 / GoldenRatio,
  impd2, PlotStyle → linstl[Black, {None, dashed}, 0.6],
  Epilog → {labels2["di", " $\sigma_i^2$ ", "", {0.12, 0.08}, fs],
    Inset[legErr, Scaled[{0.78, 0.3}]]}, Evaluate[style]]
Export["/Users/heliosight/Documents/work/UmU/projects/06_coagulation/figures/
  dls_noise_models.pdf", %, ImageResolution → 600]

```

Out[]:=



Out[]:=

```

/Users/heliosight/Documents/work/UmU/projects/06_coagulation/figures/dls_noise
_models.pdf

```

Three runs of the PS sample were used to estimate the instrumental error via SD function.

```

In[ ]:= g1dp60r = Table[xySc[dp60r[[i]], Identity, Sqrt[ $\frac{\#}{\text{Max}[dp60r[[i]][[All, 2]]}$ ]] &], {i, 3}];

g1dp60sd = {g1dp60r[[1, All, 1]], g1sd =
  StandardDeviation[#] & /@ Transpose[Table[g1dp60r[[i, All, 2]], {i, 3}]]]^T;
g1sd = ReplacePart[g1sd, 2 → Mean[g1sd[[{1, 3}]]]];
tdp60 = g1dp60r[[1, All, 1]];

```

Fitting the truncated g1 triplicates by exponential decay.

```

In[ ]:= g1cut = 50;
g1model = Exp[-a t];
g1fit = FindFit[Take[#, g1cut], g1model, {A, a}, t] & /@ g1dp60r

```

Out[]:=

```

{{A → 1., a → 5203.1}, {A → 1., a → 5232.97}, {A → 1., a → 5219.97}}

```

Generating the points for calculating residuals between $g_{1\text{fit}}$ and experimental points, aka χ^2 numerator.

```
In[*]:= glfitd = glmodel /. {a → Mean[a /. glfit], A → Mean[A /. glfit]} /. {t → tdp60};
res = (Evaluate[#[[All, 2]] & /@ gldp60r] - ConstantArray[glfitd, 3]) ;
sqres = (Evaluate[#[[All, 2]] & /@ gldp60r] - ConstantArray[glfitd, 3])2 ;
Total[#] & /@ sqres
cnstgli = Round[Total[Take[#, glcut]], 0.00001] & /@ sqres
cnstgl = Around[Mean[%], StandardDeviation[%]]
```

```
Out[*]=
{0.0818346, 0.0945422, 0.0810559}
```

```
Out[*]=
{0.00055, 0.00026, 0.00037}
```

```
Out[*]=
0.00039 ± 0.00015
```

Chi squared estimates from the triplicates

```
In[*]:= chi2 = 
$$\frac{\#}{\text{ReplaceAll}[g1sd^2, 0. \rightarrow 10^{-12}]}$$
 & /@ sqres;
Round[
$$\frac{1}{\text{Length}[\#]} \text{Total}[\#, 0.1]$$
 & /@ chi2
Around[Mean[%], StandardDeviation[%]]
Round[
$$\frac{1}{\text{Length}[\#]} \text{Total}[\#, 0.1]$$
 & /@ Evaluate[Take[#, glcut] & /@ chi2]
Around[Mean[%], StandardDeviation[%]]
Round[
$$\frac{1}{\text{Length}[\#]} \text{Total}[\#, 0.1]$$
 & /@ Evaluate[Take[#, glcut + 10] & /@ chi2]
Around[Mean[%], StandardDeviation[%]]
```

```
Out[*]=
{46.9, 45.9, 45.1}
```

```
Out[*]=
46.0 ± 0.9
```

```
Out[*]=
{10.1, 5.5, 6.7}
```

```
Out[*]=
7.4 ± 2.4
```

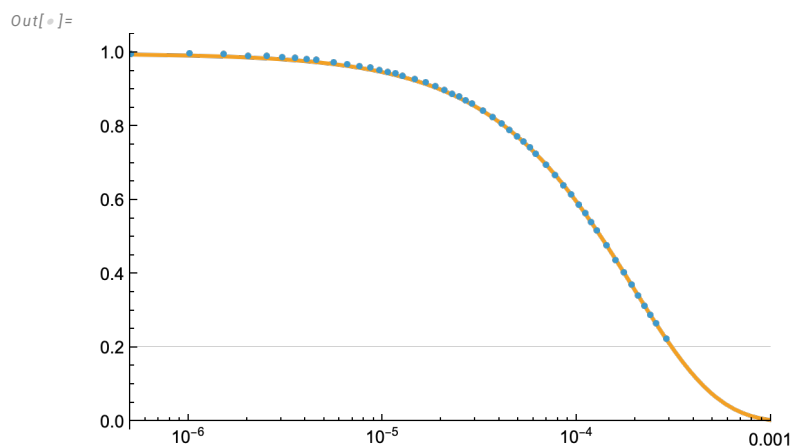
```
Out[*]=
{11.4, 7.5, 9.3}
```

```
Out[*]=
9.4 ± 2.0
```

```

In[ ]:= ListPlot[{Take[g1dp60r[[1]], g1cut], {tdp60, g1fitd}^T},
  ScalingFunctions -> {"Log"}, Joined -> {False, True},
  PlotRange -> {{0.5 × 10-6, 10-3}, All}, GridLines -> {None, {0.2}}]

```



Generating normally distributed sample with the SD of and mean of `res[[1]]`. **Non squared!!!**

```

In[ ]:= M = Length[sqres[[1]]];
σ = StandardDeviation[res[[1]]];
μ = Mean[res[[1]]];
synthetic = RandomVariate[NormalDistribution[μ, σ], M];
Combining the data, and replacing 0. for 5 10-10

```

```

In[ ]:= chidat =
  {ReplaceAll[res[[1]], 0. -> 5 × 10-10], ReplaceAll[g1sd, 0. -> 5 × 10-10], synthetic};
chidat0 = {DeleteCases[res[[1]], 0.], DeleteCases[g1sd, 0.], synthetic};
colchi = {StandardBlue, StandardRed, StandardYellow};

```

Generating indices for upper axis

```

In[ ]:= indices = Flatten[Table[{1, 2, 5} * 10^k, {k, 0, 10}]];
indices = Select[indices, # ≤ Length[tdp60] &];
picked = tdp60[[indices]];
tickslbl = {picked, indices}^T;

```

Visualization of noise and truncation

```

In[*]:= nsstr = Show[ListPlot[g1dp60sd, ScalingFunctions → {"Log10"},
  PlotMarkers → {Automatic, 4}, GridLines → {{0.0009, 2.25463}, None},
  GridLinesStyle → linstl[Black, dotted, 0.6], PlotStyle → StandardRed,
  FrameTicks → {{Automatic, Automatic}, {Automatic, tickslbl}}],
ListLinePlot[{xySc[g1dp60r[[3]], Identity, 0.02 # &],
  Take[xySc[g1dp60r[[3]], Identity, 0.2 # &], {56, -1}]],
  PlotStyle → linstl[{Black, Black}, {None, None}, 0.6],
  ScalingFunctions → {"Log10"}, PlotRange → All],
ImageSize → is, AspectRatio → 1 / GoldenRatio,
ImagePadding → {{30, 5}, {25, 25}}, PlotRange → {-0.0002, 0.023},
Epilog → {labels4["τ", s", Style["j", Italic], "g1(τ)", "", "", {0.12, 0.12}],
  Text[Style["g1×0.02", fs], Scaled[{0.1, 0.72}]],
  Text[Style["g1×0.2", fs], Scaled[{0.35, 0.82}]],
  Text[Style["σg1"], fs, StandardRed], Scaled[{0.7, 0.5}]],
  Text[Style["I", 2 fs, Gray, Bold], Scaled[{0.05, 0.92}]],
  Text[Style["II", 2 fs, Gray, Bold], Scaled[{0.48, 0.92}]], Text[
  Style["III", 2 fs, Gray, Bold], Scaled[{0.87, 0.92}]]], Evaluate[style]];

In[*]:= fdp = {tdp60, First[g1model /. g1fit /. {t → tdp60}]}T;
plrng = {{1.3 × 10-4, 0.45 × 10-2}, {-0.005, 0.5}};
nsg2g1clip = Show[ListLinePlot[Take[fdp, {42, 80}],
  ScalingFunctions → {"Log10"}, PlotStyle → linstl[Black, None, 0.6],
  PlotRange → plrng, GridLines → {{3.09 × 10-4, 7.7 × 10-4, 13.9 × 10-4}, None},
  GridLinesStyle → linstl[Black, dotted, 0.6]],
ListPlot[{Take[g1dp60r[[1]], {42, 80}], Drop[clipG2toG1[dp60r[[1]],
  20 × 10-2, 1 × 10-3, 13], 41], Take[g1dp60r[[1]], {51, 61}]],
  ScalingFunctions → {"Log10"}, GridLinesStyle → linstl[Gray, dashed, 0.6],
  PlotStyle → {Gray, Black, StandardBlue}, PlotRange → plrng,
  PlotMarkers → {Automatic, 4}], ImagePadding → {{25, 5}, {25, 25}},
ImageSize → is GoldenRatio-1, AspectRatio → 1,
Epilog → {labels2["τ", s", "g1(τ)", "", {0.13, 0.14}, fs], Text[Style["ĝ1", fs],
  Scaled[{0.32, 0.1}]], Text[Style["l1", fs], {Log[10, 3.9 × 10-4], 0.47}],
  Text[Style["l2", fs], {Log[10, 9.4 × 10-4], 0.47}],
  Text[Style["l3", fs], {Log[10, 1.7 × 10-3], 0.47}], {AbsoluteThickness[0.6],
  Gray, Line[{Log[10, 8.2 × 10-4], 0.07}, {Log[10, 0.37 × 10-2], 0.07}]]},
  Text[Style["correlator noise", fs, Gray], {Log[10, 1.7 × 10-3], 0.09}],
  Text[Style["", fs], {Log[10, 8 × 10-4], 0.19}]], Evaluate[style]];

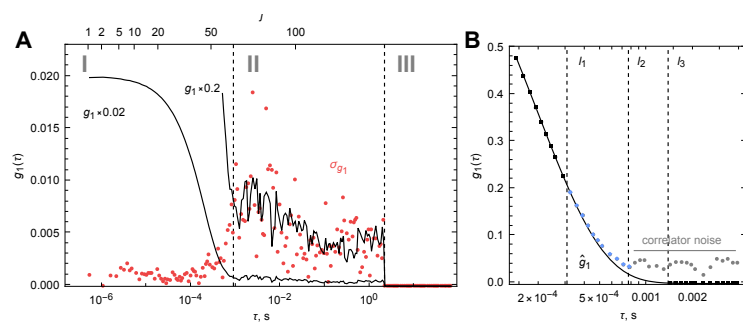
```

```

In[ ]:= Graphics[{White, EdgeForm[None],
  Rectangle[{2, 3}, {131, 55}], Inset[nsstr, {40, 30}, Automatic, 84],
  Inset[nsg2g1clip, {108, 30}, Automatic, 54],
  Text[Style["A", Bold, 2 fs, Black], {1, 53}],
  Text[Style["B", Bold, 2 fs, Black], {83, 53}]], ImageSize → 1.6 is,
ImagePadding → {{0, 0}, {0, 0}}, PlotRangeClipping → None]
Export["/Users/heliosight/Documents/work/UmU/projects/06_coagulation/figures/
  dls_noise.pdf", %, ImageResolution → 600]

```

Out[]:=



Out[]:=

```

/Users/heliosight/Documents/work/UmU/projects/06_coagulation/figures/dls_noise.
pdf

```

Noise plots

```

In[ ]:= nspts = ListPlot[chidat0, Filling → None,
  ScalingFunctions → "Log10", PlotRange → All, PlotMarkers → {Automatic, 4},
  AspectRatio → 1, PlotStyle → colchi, impd2, ImageSize →  $\frac{2}{3}$  is,
  Epilog → {labels2[Style["j", Italic], " $g_1(\tau)$ ", "", {0.1, 0.17}, fs]},
  Evaluate[style]};

nsdist = Table[Histogram[Standardize[chidat0[[i]], {-3.5, 3.5,  $\frac{7}{30}$ },
  "PDF", Axes → {True, False}, PlotRange → {{-3.5, 3.5}, {0, 1.1}},
  Frame → True, PlotRange → {10-9, 10-2}, AspectRatio → 1 / 4,
  ChartStyle → colchi[[i]], impd2, ImageSize →  $\frac{2}{3}$  is, Evaluate[style],
  Epilog → labels2[Style["z", Italic], "PDF", "", {0.36, 0.13}, fs]], {i, 3}];

autocorr = Table[{#, CorrelationFunction[chidat0[[i]], #]} & /@
  Range[Length[chidat0[[i]] - 1], {i, 3}];

autleg = PointLegend[colchi,
  {" $r_j = \hat{g}_1 - g_1$ ", " $\sigma_j = \sigma_{g_1}$ ", Style[" $e_j = N(0, \sigma_{g_1})$ ", Italic]}, LabelStyle → fs,
  LegendMarkerSize → {10, 5}, LegendLabel → Style["Samples", Bold, fs]];

nscorr = ListPlot[autocorr, PlotRange → All, PlotStyle → colchi, Filling → None,
  FillingStyle → Directive[AbsoluteThickness[0.6], Opacity[0.3]],
  PlotMarkers → {Automatic, 4}, AspectRatio → 1, impd2, ImageSize →  $\frac{2}{3}$  is,
  Epilog → {labels2[Style["j", Italic], "autocorrelation", "", {0.1, 0.17}, fs],
  Inset[autleg, Scaled[{0.75, 0.75}]]}, Evaluate[style]};

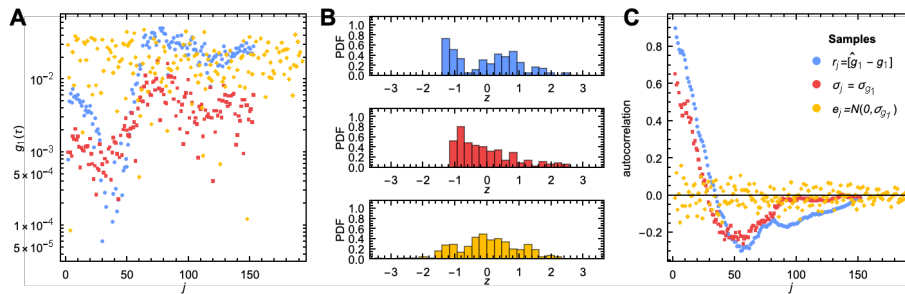
```

```

In[ ]:= Graphics[{White, EdgeForm[None], Rectangle[{2, 2}, {148, 46}],
  Inset[nspts, {25, 23}, Automatic, 50], Inset[nsdist[[1]], {75, 38}, Automatic, 48],
  Inset[nsdist[[2]], {75, 23}, Automatic, 48],
  Inset[nsdist[[3]], {75, 8}, Automatic, 48],
  Inset[nscorr, {125, 23}, Automatic, 50], Text[Style["A", Bold, 2 fs, Black],
    {1, 46}], Text[Style["B", Bold, 2 fs, Black], {52, 46}],
  Text[Style["C", Bold, 2 fs, Black], {102, 46}]], ImageSize → 2 is,
  ImagePadding → {{0, 0}, {0, 0}}, PlotRangeClipping → None]
Export["/Users/heliosight/Documents/work/UmU/projects/06_coagulation/figures/
  dls_noise_stat.pdf", %, ImageResolution → 600]

```

Out[]:=



Out[]:=

/Users/heliosight/Documents/work/UmU/projects/06_coagulation/figures/dls_noise_stat.pdf

Application of MEM to reference samples

Preprocessing the g_1 functions for constrained and unconstrained MEM. Cut at $g_1=0.2$ for constrained and $g_1 \sim 0.05$ for unconstrained MEM.

```

In[ ]:= glapfc = clipG2toG1[dapf, 16 × 10-2, 0.5 × 10-4, 13];
glbsac = clipG2toG1[dbsa, 16 × 10-2, 1 × 10-4, 13];
g1dp60cr1 = clipG2toG1[dp60r[[1]], 19 × 10-2, 9 × 10-4, 13];
g1dp60cr2 = clipG2toG1[dp60r[[2]], 19 × 10-2, 9 × 10-4, 13];
g1dp60cr3 = clipG2toG1[dp60r[[3]], 19 × 10-2, 9 × 10-4, 13];

In[ ]:= glapfu = clipG2toG1[dapf, 0.4 × 10-2, 0.5 × 10-4, 13];
glbsau = clipG2toG1[dbsa, 0.9 × 10-2, 1 × 10-4, 13];
g1dp60ur1 = clipG2toG1[dp60r[[1]], 1.4 × 10-2, 9 × 10-4, 13];
g1dp60ur2 = clipG2toG1[dp60r[[2]], 1.4 × 10-2, 9 × 10-4, 13];
g1dp60ur3 = clipG2toG1[dp60r[[3]], 1.4 × 10-2, 9 × 10-4, 13];
glumem = {glapfu, glbsau, g1dp60ur1, g1dp60ur3, g1dp60ur3};

```

Fitting with exponent

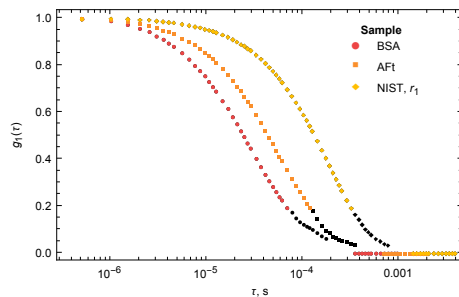
Illustration of preprocessing, left constrained, right unconstrained.

```

In[ ]:= colmem = {StandardRed, StandardOrange, StandardYellow};
legG1mem = PointLegend[colmem, {"BSA", "AFt", "NIST, r1"}, LabelStyle → fs,
  LegendMarkerSize → {15, 5}, LegendMarkers → Evaluate[{#, fs} & /@ sym],
  LegendLabel → Style["Sample", fs, Bold]];
ag1plot = Show[ListPlot[{g1bsau, g1apfu, g1dp60ur1}, ScalingFunctions → {"Log"},
  PlotRange → All, PlotStyle → Black, PlotMarkers → {Automatic, 4}],
  ListPlot[{g1bsac, g1apfc, g1dp60cr1}, ScalingFunctions → {"Log"},
  GridLines → {None, {0.05, 0.2}}, PlotStyle → colmem,
  PlotMarkers → {Automatic, 4}], impd2, AspectRatio → 1 / GoldenRatio,
  ImageSize → is, Epilog → {labels2["τ, s", "g1(τ)", "", {0.12, 0.1}, fs],
  Inset[legG1mem, Scaled[{0.8, 0.8}]]}, PlotRange → All, Evaluate[style]]

```

Out[]:=



MEM fitting of the data and exporting, the feasibility region boundary constants C_b for three samples, BSA, apo-ferritin, and NIST polystyrene standard, with three replicates.

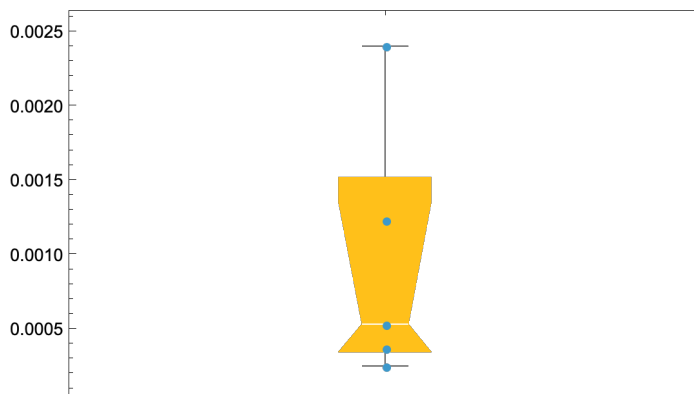
```

In[ ]:= capf = 1.0 × 0.00123;
cbsab = 1.0 × 0.0024;
cdp60r1 = 1.0 × 0.000531;
cdp60r2 = 1.0 × 0.000249;
cdp60r3 = 1.0 × 0.000369;

In[ ]:= cmem = {capf, cbsab, cdp60r1, cdp60r2, cdp60r3};
Show[BoxWhiskerChart[cmem, "Notched"], ListPlot[{1, #} & /@ cmem]]

```

Out[]:=



data import

```

In[*]:= FileNames["*csweep.h5"]
fnh5 = %[[1 ;; 5]];
Out[*]=
{dapf_csweep.h5, dbsa_csweep.h5, dp60r1_csweep.h5, dp60r2_csweep.h5,
 dp60r3_csweep.h5, ds937cut_csweep.h5, ds945cut_csweep.h5}

In[*]:= Import[fnh5[[#]], "C_constants"][[1]] & /@ Range[5]
Out[*]=
{0.00123, 0.0024, 0.000531, 0.000249, 0.000369}

In[*]:= filnm = 5;
fnh5[[filnm]]
Import[fnh5[[filnm]]]
Import[fnh5[[filnm]], "meta"]
Import[fnh5[[filnm]], "aRecovered"] // Length
Out[*]=
dp60r3_csweep.h5

Out[*]=
{/aRecovered, /gammaList, /residualVec, /meta, /resolution,
 /kernelCutoff, /C_constants, /gExperimental, /entropyVec}

Out[*]=
<|constrained → yes, glminCutoff → 0.2,
 zeroPoints → 13, date → 2026-03-12T22:20:26|>

Out[*]=
40

In[*]:= Import[fnh5[[#]], "C_constants"] & /@ Range[5] // TableForm
Out[*]//TableForm=


|          |             |             |             |             |             |
|----------|-------------|-------------|-------------|-------------|-------------|
| 0.00123  | 0.00124859  | 0.00126746  | 0.00128661  | 0.00130606  | 0.0013258   |
| 0.0024   | 0.00243627  | 0.00247309  | 0.00251047  | 0.00254841  | 0.00258692  |
| 0.000531 | 0.000539025 | 0.000547171 | 0.000555441 | 0.000563835 | 0.000572356 |
| 0.000249 | 0.000252763 | 0.000256583 | 0.000260461 | 0.000264397 | 0.000268393 |
| 0.000369 | 0.000374577 | 0.000380238 | 0.000385984 | 0.000391818 | 0.000397739 |


```

Importing the distributions for each replicate of NIST and for each constant sweep.

```

In[*]:= swd = Table[{Import[fnh5[[i]], "gammaList"]-1,  $\frac{\#}{\text{Max}[\#]}$ }T & /@
Import[fnh5[[i]], "aRecovered"], {i, 5}];

```

Calculating the diagnostic data from files

```

In[*]:= d = Table[memScanAnalysis[fnh5[[i]]], {i, Length[fnh5]};

```

The CV of NIST standard from three replicates illustrates the broadening of 17 % when maximal value is taken.

Building data for C-sweep plot.

```

In[*]:= lcurves = {d[[#, 1]], d[[#, 2, 3]]T & /@ Range[5];
knepos = Table[d[[i, 3, 1]], {i, Length[d]};
knepts = Table[lcurves[[i]][knepos[[i]]], {i, Length[d]};

```

```

In[ ]:= trueCVnist = 0.134;
Table[d[[i, 2, 3]][[knepos[[i]]], {i, 3, 5}]
Around[Mean[%], StandardDeviation[%]]
Max[%]
broadening = Round[1 - % / trueCVnist, 0.01]

Out[ ]:=
{0.111355, 0.0923616, 0.0714502}

Out[ ]:=
0.092 ± 0.020

Out[ ]:=
0.111355

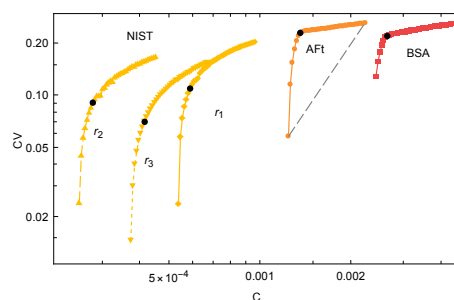
Out[ ]:=
0.17

In[ ]:= colkne = Join[{StandardOrange, StandardRed}, ConstantArray[StandardYellow, 3]];

In[ ]:= tcrd = {{0.00335444, 0.176516}, {0.001525, 0.185075},
  {0.000735161, 0.0805323}, {0.000429798, 0.0421645}, {0.000292296, 0.062213}};
tsmp = {"BSA", "Aft", "r1", "r3", "r2"};
clabl = Table[Text[Style[tsmp[[i]], fs], Log10@tcrd[[i]]], {i, Length[tcrd]}];
alcplot = Show[ListLinePlot[lcurves, PlotRange → All,
  ScalingFunctions → {"Log10", "Log10"}, PlotRange → All, PlotStyle →
  linstl[colkne, Join[ConstantArray[None, 3], {dashed, dotted}], 0.6],
  PlotMarkers → {Automatic, 5}], ListPlot[knepts, PlotRange → All,
  ScalingFunctions → {"Log10", "Log10"}, PlotStyle → Black,
  PlotMarkers → {Automatic, 6}], PlotRange → All,
  impd2, AspectRatio → 1 / GoldenRatio, ImageSize → is,
  Epilog → {labels2["C", "CV", "", {0.13, 0.09}, fs],
    {linstl[Gray, dashed, 0.6], Line[Log10@lcurves[[1, {1, -1}]]],
    Text[Style["NIST", fs], Log10@{4 × 10-4, 0.22}], clabl}, Evaluate[style]]

Out[ ]:=

```



First and second peaks of constrained mem sweep and their amplitude and SD for BSA of first three runs

```

In[ ]:= peak1H = Last@Last[SortBy[swd[[#, 1]], Last]] & /@ Range[3];
peak2H = Last@Last[SortBy[Take[swd[[#, 1], 150], Last]] & /@ Range[2]
StandardDeviation[Last@Last[SortBy[Take[swd[[1, #]], 150], Last]] & /@ Range[3]];
dbsa41 = Around[%%[[2], %]

Out[ ]:=
{0.31154, 0.326192}

Out[ ]:=
0.326 ± 0.024

In[ ]:= peak1 = First@Last[SortBy[swd[[#, 1]], Last]] & /@ Range[3]
peak2 = First@Last[SortBy[Take[swd[[#, 1], 150], Last]] & /@ Range[2]

Out[ ]:=
{0.0000474692, 0.0000241384, 0.000191182}

Out[ ]:=
{0.000160787, 0.0000903602}

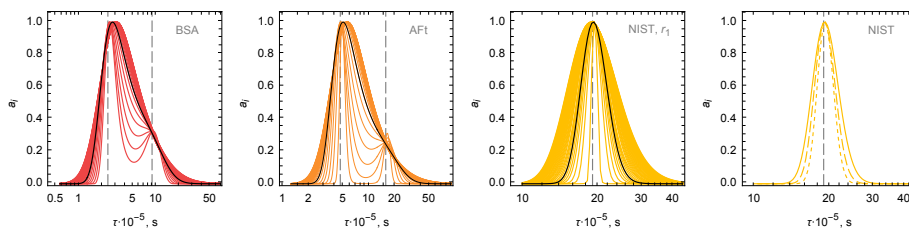
```

```

In[ ]:= {ListLinePlot[
  Take[xySc[#, 105 # &], {20, -120}] & /@ Append[swd[[2]], swd[[2]][[knepos[[2]]]],
  ScalingFunctions → {"Log"}, GridLines → {105 {peak1[[2]], peak2[[2]]}, None},
  GridLinesStyle → linstl[Gray, dashed, 0.6],
  PlotStyle → linstl[Append[ConstantArray[StandardRed, 40], Black], None, 0.6],
  PlotRange → {All, All}, AspectRatio → 1, Impd2, ImageSize →  $\frac{1}{2}$  is,
  Epilog → {labels2[" $\tau \cdot 10^{-5}$ , s", "ai", "", {0.2, 0.2}, fs],
    Text[Style["BSA", fs, Gray], Scaled[{0.8, 0.9}]]}, Evaluate[style]},
ListLinePlot[Take[xySc[#, 105 # &], {40, -160}] & /@
  Append[swd[[1]], swd[[1]][[knepos[[1]]]], ScalingFunctions → {"Log"}, PlotStyle →
    linstl[Append[ConstantArray[StandardOrange, 40], Black], None, 0.5],
  PlotRange → {All, All}, PlotRange → All,
  GridLines → {105 {peak1[[1]], peak2[[1]]}, None},
  GridLinesStyle → linstl[Gray, dashed, 0.6], AspectRatio → 1, Impd2,
  ImageSize →  $\frac{1}{2}$  is, Epilog → {labels2[" $\tau \cdot 10^{-5}$ , s", "ai", "", {0.2, 0.2}, fs],
    Text[Style["AFt", fs, Gray], Scaled[{0.8, 0.9}]]}, Evaluate[style]},
ListLinePlot[
  Take[xySc[#, 105 # &], {110, -265}] & /@ Append[swd[[3]], swd[[3]][[knepos[[3]]]],
  ScalingFunctions → {"Log"}, GridLines → {105 {peak1[[3]]}, None},
  GridLinesStyle → linstl[Gray, dashed, 0.6], PlotStyle →
    linstl[Append[ConstantArray[StandardYellow, 40], Black], None, 0.6],
  PlotRange → {All, All}, AspectRatio → 1, Impd2, ImageSize →  $\frac{1}{2}$  is,
  Epilog → {labels2[" $\tau \cdot 10^{-5}$ , s", "ai", "", {0.2, 0.2}, fs],
    Text[Style["NIST, r1", fs, Gray], Scaled[{0.8, 0.9}]]}, Evaluate[style]},
ListLinePlot[Take[xySc[#, 105 # &], {110, -265}] & /@
  Table[swd[[i]][[knepos[[i]]]], {i, 3, 5}], ScalingFunctions → {"Log"},
  GridLines → {105 {peak1[[3]]}, None}, GridLinesStyle → linstl[Gray, dashed, 0.6],
  PlotStyle → linstl[StandardYellow, {None, dashed, dotted}, 0.6],
  PlotRange → {All, All}, AspectRatio → 1, Impd2, ImageSize →  $\frac{1}{2}$  is,
  Epilog → {labels2[" $\tau \cdot 10^{-5}$ , s", "ai", "", {0.2, 0.2}, fs],
    Text[Style["NIST", fs, Gray], Scaled[{0.8, 0.9}]]}, Evaluate[style]}}];
aswplot = GraphicsRow[sh[sh] & /@ %, Spacings → 0, ImageSize → 2 is]

```

Out[]:=



The distributions selected from the sweep via knee of L - curve. Two versions are the constrained and unconstrained MEM.

NB. This computation needs the data from mp_calibration.nb

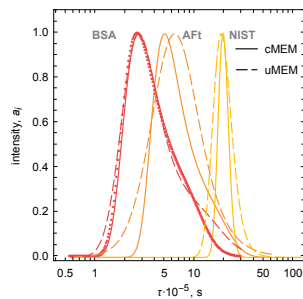
```

In[ ]:= cmsd = Table[swd[[i]][knepos[[i]]], {i, 5}];
cmspdf = {#^T[[1]], Rescale[#^T[[2]]]^T & /@ Evaluate[pdfLinear[#] & /@ cmsd];

In[ ]:= legpdf = LineLegend[linstl[Black, {None, dashed}, 0.6],
  {"cMEM", "uMEM"}, LabelStyle → fs, LegendMarkerSize → {13, 5}];
acplot = Show[ListLinePlot[Take[xySc[#, 10^5 # &], {60, -120}] & /@ cmsd[[1 ;; 3]],
  ScalingFunctions → {"Log"}, PlotMarkers → {None, {sym[[1]], 2}, None},
  PlotStyle → linstl[colkne, None, 0.6], PlotRange → All],
ListLinePlot[Take[xySc[#, 10^5 # &], {60, -120}] & /@ umsd[[1 ;; 3]],
  ScalingFunctions → {"Log"}, Filling → None, PlotStyle →
  linstl[{StandardOrange, StandardRed, StandardYellow}, dashed, 0.6],
  PlotRange → All], PlotRange → {All, {All, 1.1}},
  AspectRatio → 1, Impd2, ImageSize →  $\frac{2}{3}$  is,
  Epilog → {labels2[" $\tau \cdot 10^{-5}$ ", s", "intensity,  $a_i$ ", "", {0.12, 0.15}, fs],
    Text[Style["AFt", fs, Gray, Bold], Scaled[{0.55, 0.9}]],
    Text[Style["BSA", fs, Gray, Bold], Scaled[{0.2, 0.9}]],
    Text[Style["NIST", fs, Gray, Bold], Scaled[{0.75, 0.9}]],
    Inset[legpdf, Scaled[{0.845, 0.8}]]}, Evaluate[style]]

```

Out[]:=



```

In[ ]:= fnAll = FileNames["*lsweep.h5"]
Import[fnAll[[1]], "meta"]
Import[fnAll[[1]], "samples"] // Keys
ums = Import[fnAll[[1]], "samples"];

```

Out[]:=

```
{dall_lsweep.h5}
```

Out[]:=

```

<|constrained → no, glminCutoff → 0.05, zeroPoints → 13,
  date → 2026-03-14T15:16:54, resolution → 0.02, kernelCutoff → 50|>

```

Out[]:=

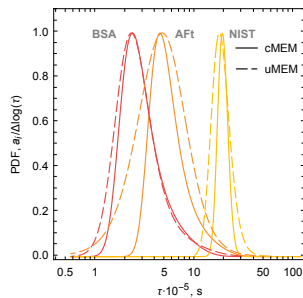
```
{AFt, BSA, NISTr1, NISTr2, NISTr3}
```

```

In[*]:= umsd = Table[{ums[[i]]["gammaList"]^-1,  $\frac{\text{ums}[[i]]["aRecovered"]}{\text{Max}[\text{ums}[[i]]["aRecovered"]]}$ }^T,
  {i, Range[Length[ums]]}];
umspdf = {#^T[[1]], Rescale[#^T[[2]]]}^T & /@ Evaluate[pdfLinear[#] & /@ umsd];
In[*]:= auplot = Show[ListLinePlot[Take[xySc[#, 10^5 # &], {60, -120}] & /@ cmspdf[[1 ;; 3]],
  ScalingFunctions -> {"Log"}, PlotMarkers -> {None, None, None},
  PlotStyle -> linstl[colkne, None, 0.6], PlotRange -> All],
ListLinePlot[Take[xySc[#, 10^5 # &], {60, -120}] & /@ umspdf[[1 ;; 3]],
  ScalingFunctions -> {"Log"}, PlotStyle ->
  linstl[{StandardOrange, StandardRed, StandardYellow}, dashed, 0.6],
  PlotRange -> All, Filling -> None], PlotRange -> {All, {All, 1.1}},
  AspectRatio -> 1, Impd2, ImageSize ->  $\frac{2}{3}$  is,
  Epilog -> {labels2[" $\tau \cdot 10^{-5}$ , s", "PDF,  $a_i / \Delta \log(\tau)$ ", "", {0.12, 0.15}, fs],
    Text[Style["AFt", fs, Gray, Bold], Scaled[{0.52, 0.9}]],
    Text[Style["BSA", fs, Gray, Bold], Scaled[{0.2, 0.9}]],
    Text[Style["NIST", fs, Gray, Bold], Scaled[{0.75, 0.9}]],
    Inset[legpdf, Scaled[{0.845, 0.8}]]}, Evaluate[style]]

```

Out[*]=



Peak positions of constrained MEM inversion

```

In[*]:= 10^5 SortBy[#, Last][[-1, 1]] & /@ cmspdf
TauToRadiusNM[10^-5 * %]

```

Out[*]=

```
{4.38196, 2.31919, 18.7397, 18.7397, 19.1182}
```

Out[*]=

```
{7.5, 4., 31.9, 31.9, 32.6}
```

Peak positions of unconstrained MEM inversion

```

In[*]:= 10^5 SortBy[#, Last][[-1, 1]] & /@ umspdf
TauToRadiusNM[10^-5 * %]

```

Out[*]=

```
{4.65292, 2.27327, 17.6484, 17.6484, 18.0049}
```

Out[*]=

```
{7.9, 3.9, 30.1, 30.1, 30.7}
```

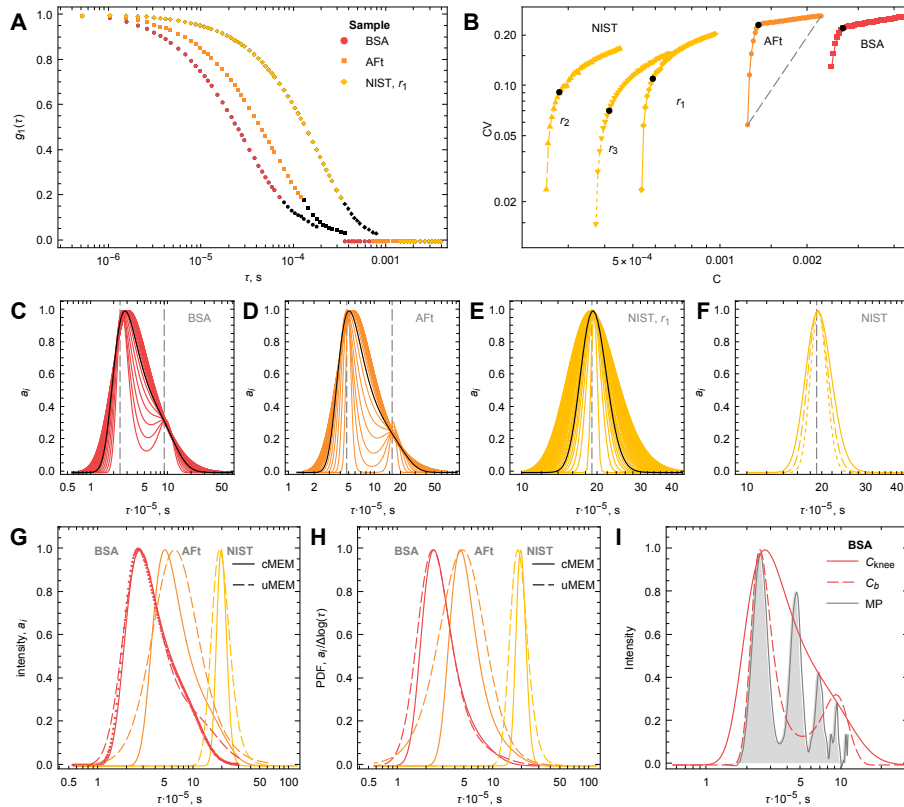
Illustration

```

In[ ]:= Graphics[{White, EdgeForm[None],
  Rectangle[{2, 2}, {158, 140}], Inset[acplot, {26, 26}, Automatic, 53],
  Inset[auplot, {26 + 53, 26}, Automatic, 53],
  Inset[mpbsapl, {26 + 54.5 + 53, 26}, Automatic, 53],
  Inset[aglplot, {39, 118}, Automatic, 79],
  Inset[alclplot, {41 + 80, 118}, Automatic, 79],
  Inset[aswplot, {80, 72}, Automatic, 159],
  Text[Style["A", Bold, 2 fs, Black], {1, 140}],
  Text[Style["B", Bold, 2 fs, Black], {81, 140}],
  Text[Style["C", Bold, 2 fs, Black], {1, 89}],
  Text[Style["D", Bold, 2 fs, Black], {42, 89}],
  Text[Style["E", Bold, 2 fs, Black], {82, 89}],
  Text[Style["F", Bold, 2 fs, Black], {122, 89}],
  Text[Style["G", Bold, 2 fs, Black], {1, 49}],
  Text[Style["H", Bold, 2 fs, Black], {54, 49}],
  Text[Style["I", Bold, 2 fs, Black], {54 + 53, 49}]], ImageSize → 2 is,
ImagePadding → {{0, 0}, {0, 0}}, PlotRangeClipping → None]
Export["/Users/heliosight/Documents/work/UmU/projects/06_coagulation/figures/
  dls_application.pdf", %, ImageResolution → 600]

```

Out[]:=



Out[]:=

```

/Users/heliosight/Documents/work/UmU/projects/06_coagulation/figures/
  dls_application.pdf

```

Summary table

Sweep constants

```

In[ ]:= cconst = Import[fnh5[[#]], "C_constants"] & /@ Range[5];
cbconst = First[#] & /@ cconst
ckconst = MapThread[Part, {cconst, knepos}]

Out[ ]:=
{0.00123, 0.0024, 0.000531, 0.000249, 0.000369}

Out[ ]:=
{0.00134583, 0.00262602, 0.000581007, 0.000276567, 0.000409852}

```

Lambda constants

```

In[ ]:= lambda = ums[[#]]["lambda"] & /@ Range[Length[ums]]

Out[ ]:=
{0.0421, 0.0121, 0.00135, 0.00039, 0.00023}

```

Peak positions left to right in 10^{-5} s

```

In[ ]:= TableForm[{cbconst, ckconst, lambda,
  peakc = Last[SortBy[cmSPdf[[#]], Last]][[1]] & /@ Range[5];
  TauToRadiusNM[peakc],
  Round[105 peakc, 0.01],
  peaku = Last[SortBy[umSPdf[[#]], Last]][[1]] & /@ Range[5];
  TauToRadiusNM[peaku],
  Round[105 peaku, 0.01]},
  TableHeadings → {{"Cb", "Ck", "λk", "Rc, nm", "τc, 10-5 s", "Ru, nm", "τu, 10-5 s"},
    {"Aft", "BSA", "r1", "r3", "r2"}}]

```

	Aft	BSA	r ₁	r ₃	r ₂
C _b	0.00123	0.0024	0.000531	0.000249	0.000369
C _k	0.00134583	0.00262602	0.000581007	0.000276567	0.000409852
λ _k	0.0421	0.0121	0.00135	0.00039	0.00023
R _c , nm	7.5	4.	31.9	31.9	32.6
τ _c , 10 ⁻⁵ s	4.38	2.32	18.74	18.74	19.12
R _u , nm	7.9	3.9	30.1	30.1	30.7
τ _u , 10 ⁻⁵ s	4.65	2.27	17.65	17.65	18.

Entropy

Calculating Shannon's entropy for gaussian and power law distributions

```

In[ ]:= ClearAll[shannon, probGaussian, probPower]

shannon[a_] := -Total[Replace[a Log[a], Indeterminate → 0, {1}]]

x = Range[1, 100, 0.1];
dx = x[[2]] - x[[1]];

probGaussian[mu_, sd_] :=
Module[{logp, p, a}, logp = (-1/2 Log[2 Pi sd^2] - (# - mu)^2 / (2 sd^2)) & /@ x;
p = Exp[logp];
a = p dx;
a = a / Total[a];
a]

probPower[alpha_] := Module[{p, a}, p = x^-alpha;
a = p dx;
a / Total[a]]

Standard deviation scanning range

In[ ]:= sdr = Range[0.5, 10, 0.5];

entropyGaussian = shannon /@ (probGaussian[50, #] & /@ sdr);

entropyPower15 = shannon[probPower[1.5]];
entropyPower18 = shannon[probPower[1.8]];

```

```

In[ ]:= tau192 = dbSA[All, 1];
g1Totaln =
  {#, If[g1Total[#] < 10-12, 0, g1Total[#] + RandomReal[{-10-2, 10-2}] & /@ tau192];
Show[ListPlot[{g1Totaln, Take[g1Totaln, 65]}, ScalingFunctions → {"Log"}],
  Plot[g1Total[t], {t, 0.5 × 10-6, 0.01}, ScalingFunctions → {"Log"}]]

```

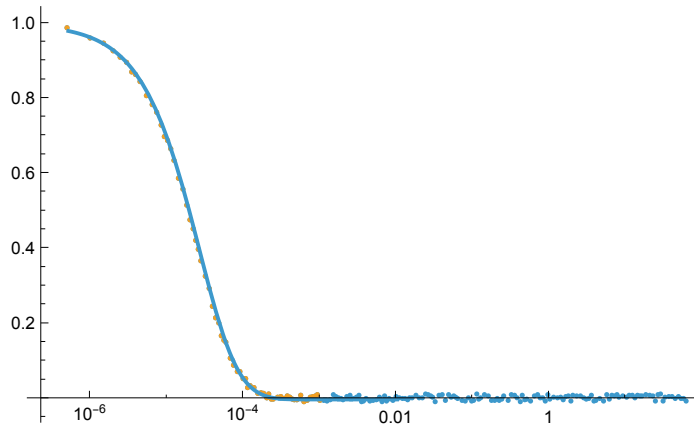
General: Exp[−717.766] is too small to represent as a normalized machine number; precision may be lost. [i](#)

General: Exp[−772.978] is too small to represent as a normalized machine number; precision may be lost. [i](#)

General: Exp[−823.172] is too small to represent as a normalized machine number; precision may be lost. [i](#)

General: Further output of General::munfl will be suppressed during this calculation. [i](#)

Out[]:=



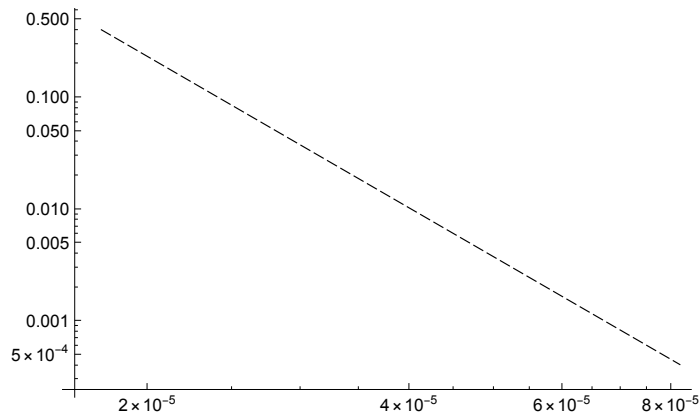
The synthetic distribution in time domain is generated via converting the hydrodynamic radius to relaxation time using Stokes-Einstein and wave vector

```

In[ ]:= synpwr = {RadiusNMTToTau[109 Rh[#] & /@ Range[100]], weights}^T;
ListLinePlot[synpwr, ScalingFunctions → {"Log", "Log"},
  PlotStyle → {Instl[Black, dashed, 0.6]}]

```

Out[]:=



Running unconstrained MEM with wide range of regularisation parameter λ $10^{-2} \dots 10^{-6}$. Converting the output into time domain and from log into linear space.

```
In[ ]:= Timing[gpwr = dlsMEMu[Take[g1Totaln, 65], 0.05, 5 × 10-#] & /@ Range[2, 6];]
rdst = {#[[1]]-1, #[[2]]}ᵀ & /@ gpwr;
rdstpdf = pdfLinear[#] & /@ rdst;
```

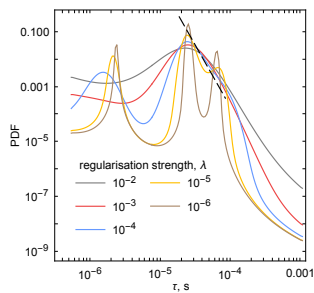
```
Out[ ]:=
{34.7045, Null}
```

```
In[ ]:= colpwr = {Gray, StandardRed, StandardBlue, StandardYellow, StandardBrown}
txtpwr = {"10-2", "10-3", "10-4", "10-5", "10-6"};
legpwr = LineLegend[linstl[colpwr, None, 0.6],
  txtpwr, LabelStyle → fs, LegendMarkerSize → {17, 5}, LegendLabel →
  Style["regularisation strength, λ", fs], LegendLayout → {"Column", 2}];
```

```
Out[ ]:=
{■, ■, ■, ■, ■}
```

```
In[ ]:= mempwr =
Show[ListLinePlot[xySc[#, Identity, 10-6 # &] & /@ rdstpdf, ScalingFunctions →
  {"Log", "Log"}, PlotRange → All, PlotStyle → linstl[colpwr, None, 0.6]],
ListLinePlot[synpwr, ScalingFunctions → {"Log", "Log"},
  PlotStyle → linstl[Black, dashed, 0.6]], PlotRange → All,
PlotStyle → Join[ConstantArray[Gray, 7], {StandardRed, StandardBlue}],
PlotMarkers → {Automatic, 3}, PlotRange → All, ImageSize →  $\frac{2}{3}$  is, AspectRatio → 1,
Evaluate@impd2, Epilog → {labels2["τ, s", "PDF", "", {0.1, 0.16}, fs],
  Inset[legpwr, Scaled[{0.35, 0.25}]]}, Evaluate[style]
```

```
Out[ ]:=
```

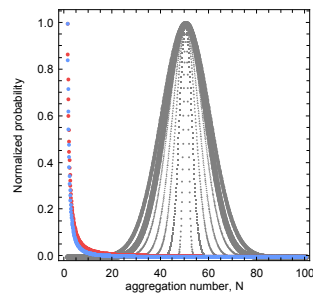


```

In[ ]:= ndpln = ListPlot[
  Join[Evaluate[{{x,  $\frac{\text{probGaussian}[50, \#]}{\text{Max}[\text{probGaussian}[50, \#]]^1}$ }T & /@ thin[Drop[sdr, 1], 3]},
    {{x,  $\frac{\text{probPower}[1.5]}{\text{Max}[\text{probPower}[1.5]]}$ }T, {x,  $\frac{\text{probPower}[1.8]}{\text{Max}[\text{probPower}[1.8]]}$ }T}}, PlotRange → All,
  PlotStyle → Join[ConstantArray[Gray, 7], {StandardRed, StandardBlue}],
  PlotMarkers → {Automatic, 2.5}, PlotRange → All, ImageSize →  $\frac{2}{3}$  is,
  AspectRatio → 1, Evaluate@impd2, Epilog → {labels2["aggregation number, N",
    "Normalized probability", "", {0.1, 0.16}, fs]}, Evaluate[style]}

```

Out[]:=



```

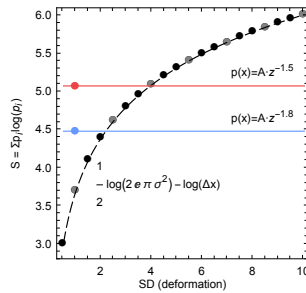
In[ ]:= cols = {Black, StandardRed, StandardBlue};

tfS = Style[TraditionalForm[ $\frac{1}{2} \text{Log}[2 \pi e \sigma^2] - \text{Log}[\Delta x]$ ], fs];

Spl = Show[ListPlot[{Transpose[{sdr, entropyGaussian}],
  {{sdr[[2]], entropyPower15}}, {{sdr[[2]], entropyPower18}}},
  PlotMarkers → {sym[[1]], fs}, PlotRange → All, PlotStyle → cols],
  ListPlot[thin[Drop[Transpose[{sdr, entropyGaussian}], 1], 3],
  PlotMarkers → {sym[[1]], fs}, PlotRange → All, PlotStyle → Gray],
  Plot[ $\frac{1}{2} \text{Log}[2 \pi e \sigma^2] - \text{Log}[dx]$ , entropyPower15, entropyPower18], {σ, 0.5, 10},
  PlotRange → All, PlotStyle → linstl[cols, {dashed, None, None}, 0.6]],
  AspectRatio → 1, Impd2, ImageSize →  $\frac{2}{3}$  is, PlotRange → {{0.3, 10.2}, All},
  Epilog → {labels2["SD (deformation)", "S =  $\sum p_i \log(p_i)$ ", "", {0.1, 0.16}, fs],
    Text[Style[tfS, fs], Scaled[{0.4, 0.3}]],
    Text[Style["p(x)=A·z-1.5", fs], Scaled[{0.82, 0.75}]],
    Text[Style["p(x)=A·z-1.8", fs], Scaled[{0.82, 0.565}]]], Evaluate[style]]

```

Out[]:=

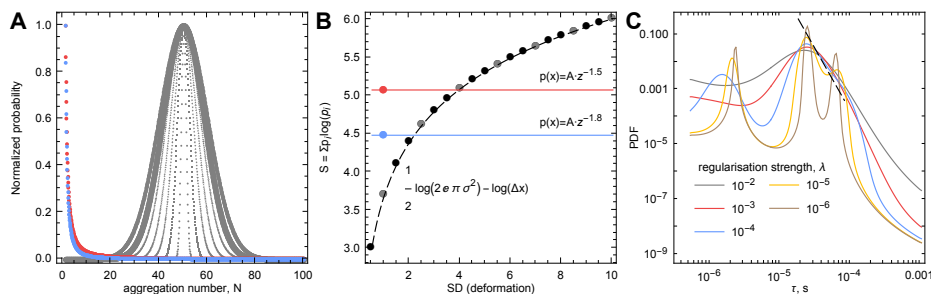


```

In[ ]:= GraphicsGrid[{{ndpln, Spl, mempwr}}, ImageSize → 2 is, Spacings → 0,
  Epilog → {Text[Style["A", Bold, 2 fs, Black], Scaled[{0.01, 0.94}]],
    Text[Style["B", Bold, 2 fs, Black], Scaled[{0.34, 0.94}]],
    Text[Style["C", Bold, 2 fs, Black], Scaled[{0.675, 0.94}]]}],
  Export["/Users/heliosight/Documents/work/UmU/projects/06_coagulation/figures/
    mem_entropy.pdf", %, ImageResolution → 600]

```

Out[]:=



Out[]:=

```

/Users/heliosight/Documents/work/UmU/projects/06_coagulation/figures/
  mem_entropy.pdf

```